

Element C. Presentation and Justification of Solution Requirements

For the original design that we are coming up with we are taking the outline of the base design and modifying the shape of the 9 inch tall bookend and changing it to a different shape and making it out of recycled material to cost less but also to hopefully reduce the material needed to use it. Changing the design to a triangular shape can keep the sturdy structure of the previous bookend while using less material and hopefully holding up the same or more weight.

List of most important Criteria:

- Made out of recycled materials - highest priority because the main goal of creating the bookends is to be out of recycled materials so that it is inexpensive.
- Durability/Survivability - The book ends aren't efficient if they cannot last a long time.
- Sturdy/Strong - The bookends need to be able to hold a certain amount of weight so their strength is important.
- Movable - The librarian must be able to move the bookends from shelf to shelf for her desired arrangement.
- Space Consumption - While a bottom priority because everything else is more important it is still a goal to take up as little space on the bookshelf as possible so that more books can be placed on it to maximize the space.

	Great	Acceptable	Not well	Nothing achieved
Moveable	Can be moved from bookshelf to bookshelf easily.		Isn't very movable and is stuck in place, but with effort can be moved.	Can not be moved.
Sturdy	Can hold as many books as needed close to holding an entire row of books on a bookshelf by itself.	Can hold about half a bookshelves worth of books in the library.	Can hold about 10 medium size books without falling over but crumbles when having to hold over 2 large books.	Can hold little to no weight
Durable	Can last for a long time and can be used multiple times	Has an ok stability and can last for a couple months before	Only last for a couple weeks before it breaks down and needs	Can only last for one use before it breaks.

	with no wear or tear from the books. Can last for at least an entire school year.	wear and tear starts to affect its performance.	to be rebuilt/replaced.	
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Primary Stakeholder Validation: These design requirements will meet the needs of the Primary stakeholder (Ms. Tatro), because when built, this will provide a bookend similar to the expensive ones she has already purchased, as they will be durable, sturdy, and moveable, creating a seamless transition to a more organized and put together library.